conceptbox colback=bglight, colframe=primary, arc=4pt, outer arc=4pt, boxrule=1.5pt, left=12pt, right=12pt, top=10pt, bottom=10pt

Exercise 1C Complete Solutions

Topic: Multiplication, Division, and Sequential Properties of Integers

This reference guide provides fully worked out, step-by-step solutions for every problem in **Exercise 1C**. Each solution includes the relevant algebraic rules, properties, and logical steps.

Key Arithmetic Sign Rules for Integers

Multiplication and Division rules:

- **Like Signs:** The product or quotient of two integers with the **same sign** is always **positive**.

$$(+) \times (+) = + \quad \text{and} \quad (-) \times (-) = +$$
$$(+) \div (+) = + \quad \text{and} \quad (-) \div (-) = +$$

- **Unlike Signs:** The product or quotient of two integers with **unlike signs** is always **negative**.

$$(+) \times (-) = - \quad \text{and} \quad (-) \times (+) = -$$
$$(+) \div (-) = - \quad \text{and} \quad (-) \div (+) = -$$

1. Find the product:

(i) $\mathbf{18 \times 4}$
Both integers are positive (like signs), so the product is positive.
$18 \times 4 = \mathbf{72}$

(ii) $(\mathbf{-25}) \times \mathbf{6}$
One integer is negative and one is positive (unlike signs), so the product is negative.
$(-25) \times 6 = -(25 \times 6) = \mathbf{-150}$

(iii) $(\mathbf{-30}) \times \mathbf{7}$
Unlike signs, so the product is negative.
$(-30) \times 7 = -(30 \times 7) = \mathbf{-210}$

(iv) $\mathbf{8} \times (\mathbf{-15})$
Unlike signs, so the product is negative.
$8 \times (-15) = -(8 \times 15) = \mathbf{-120}$

(v) $\mathbf{20} \times (\mathbf{-10})$
Unlike signs, so the product is negative.
$20 \times (-10) = -(20 \times 10) = \mathbf{-200}$

(vi) $(\mathbf{-12}) \times (\mathbf{-15})$
Both integers are negative (like signs), so the product is positive.
$(-12) \times (-15) = +(12 \times 15) = \mathbf{180}$

(vii) $(\mathbf{-8}) \times (\mathbf{-13})$
Like signs, so the product is positive.
$(-8) \times (-13) = +(8 \times 13) = \mathbf{104}$

(viii) $(\mathbf{-20}) \times (\mathbf{-1})$
Like signs, so the product is positive.
$(-20) \times (-1) = +(20 \times 1) = \mathbf{20}$

(ix) $(\mathbf{-9}) \times \mathbf{0}$
The multiplicative property of zero states that any integer multiplied by 0 is 0.
$(-9) \times 0 = \mathbf{0}$

(x) $\mathbf{0} \times (\mathbf{-11})$
Any integer multiplied by 0 is 0.
$0 \times (-11) = \mathbf{0}$

2. Find the product:

(i) $\{(\mathbf{-9}) \times \mathbf{8}\} \times (\mathbf{-5})$
First, solve the brackets: $(-9) \times 8 = -72$.
Then, multiply by the third integer: $(-72) \times (-5) = \mathbf{360}$ (like signs yield positive).

(ii) $\{(\mathbf{-10}) \times (\mathbf{-5})\} \times \mathbf{6}$
First, solve the brackets: $(-10) \times (-5) = 50$.
Then, multiply by the third integer: $50 \times 6 = \mathbf{300}$.

(iii) $\{(\mathbf{-12}) \times (\mathbf{-15})\} \times (\mathbf{-2})$
First, solve the brackets: $(-12) \times (-15) = 180$.
Then, multiply by the third integer: $180 \times (-2) = \mathbf{-360}$.

(iv) $(\mathbf{-8}) \times \{(\mathbf{-5}) \times (\mathbf{-3})\}$
First, solve the brackets: $(-5) \times (-3) = 15$.
Then, multiply: $(-8) \times 15 = \mathbf{-120}$.

(v) $(\mathbf{-11}) \times \{(\mathbf{-8}) \times \mathbf{5}\}$
First, solve the brackets: $(-8) \times 5 = -40$.
Then, multiply: $(-11) \times (-40) = \mathbf{440}$.

3. Verify the following:

(i) $(-\mathbf{14}) \times (-\mathbf{8}) = (-\mathbf{8}) \times (-\mathbf{14})$
L.H.S.: $(-14) \times (-8) = 112$
R.H.S.: $(-8) \times (-14) = 112$
Since **L.H.S. = R.H.S. = 112**, the equation is verified.
Property shown: Commutative Law of Multiplication ($a \times b = b \times a$).

(ii) $\{(-\mathbf{7}) \times \mathbf{5}\} \times (-\mathbf{6}) = (-\mathbf{7}) \times \{\mathbf{5} \times (-\mathbf{6})\}$
L.H.S.: $\{(-7) \times 5\} \times (-6) = (-35) \times (-6) = 210$
R.H.S.: $(-7) \times \{5 \times (-6)\} = (-7) \times (-30) = 210$
Since **L.H.S. = R.H.S. = 210**, the equation is verified.
Property shown: Associative Law of Multiplication ($(a \times b) \times c = a \times (b \times c)$).

(iii) $(-\mathbf{10}) \times \{(-\mathbf{7}) + (-\mathbf{9})\} = \{(-\mathbf{10}) \times (-\mathbf{7})\} + \{(-\mathbf{10}) \times (-\mathbf{9})\}$
L.H.S.: $(-10) \times \{-7 + (-9)\} = (-10) \times (-16) = 160$
R.H.S.: $\{(-10) \times (-7)\} + \{(-10) \times (-9)\} = 70 + 90 = 160$
Since **L.H.S. = R.H.S. = 160**, the equation is verified.
Property shown: Distributive Law of Multiplication over Addition ($a \times (b+c) = a \times b + a \times c$).

4. Find the quotient:

(i) $\mathbf{28} \div (-\mathbf{7})$
Unlike signs, quotient is negative.
$28 \div (-7) = -\mathbf{4}$

(ii) $(-\mathbf{65}) \div \mathbf{13}$
Unlike signs, quotient is negative.
$(-65) \div 13 = -\mathbf{5}$

(iii) $(-\mathbf{66}) \div (-\mathbf{6})$
Like signs, quotient is positive.
$(-66) \div (-6) = \mathbf{11}$

(iv) $(-\mathbf{9}) \div (-\mathbf{1})$
Like signs, quotient is positive.
$(-9) \div (-1) = \mathbf{9}$

(v) $\mathbf{0} \div (-\mathbf{2})$
Dividing zero by any non-zero integer results in zero ($0 \div a = 0$).
$0 \div (-2) = \mathbf{0}$

(vi) $(-\mathbf{12}) \div (-\mathbf{12})$
Any non-zero integer divided by itself is 1 ($a \div a = 1$).
$(-12) \div (-12) = \mathbf{1}$

5. Write all even integers between:

(i) (-4) **and 11**
The even integers strictly between -4 and 11 are:
$-2, 0, 2, 4, 6, 8, 10$

(ii) (-13) **and** (-7)
The even integers strictly between -13 and -7 are:
$-12, -10, -8$

6. Write all odd integers between:

(i) (-1) **and 7**
The odd integers strictly between -1 and 7 are:
$1, 3, 5$

(ii) (-20) **and** (-14)
The odd integers strictly between -20 and -14 are:
$-19, -17, -15$

7. Write five consecutive even integers succeeding -21:

Succeeding -21 means finding even integers strictly greater than -21 in increasing order.
The first even integer greater than -21 is -20.
Adding 2 successively to find the next four even integers:

- $-20 + 2 = -18$
- $-18 + 2 = -16$
- $-16 + 2 = -14$
- $-14 + 2 = -12$

Thus, the five consecutive even integers are: $-20, -18, -16, -14, -12$.

8. Write five consecutive odd integers preceding -36:

Preceding -36 means finding odd integers strictly less than -36 in decreasing order.
The first odd integer smaller than -36 is -37.
Subtracting 2 successively to find the next four odd integers:

- $-37 - 2 = -39$
- $-39 - 2 = -41$
- $-41 - 2 = -43$
- $-43 - 2 = -45$

Thus, the five consecutive odd integers are: $-37, -39, -41, -43, -45$.

9. The product of two integers is -120. If one number is 15, find the other:

Let the other integer be x.
We are given:

$$15 \times x = -120$$

To find x, we divide both sides by 15:

$$x = \frac{-120}{15}$$

Using the unlike signs division rule, the quotient is negative:

$$x = -(120 \div 15) = -8$$

The other integer is -8.